

# Build your cardboard cockpit

Two paper markers. One webcam. Flight controls you can make by hand.

**Gather:** corrugated cardboard, shoebox, tape, capped marker pen, scissors, printed markers, webcam, desk and chair. Allow about 45-60 minutes. Round exposed corners.

## 01 CUT THE PARTS

### YOKE BLANKS

2 pieces, 32 x 20 cm. Remove a centered 16 x 12 cm opening from the top of each. Handles and lower bridge remain 8 cm wide.

### YOKE MARKER PLATE

1 piece, 11 x 11 cm. Mount centered on the camera-facing side.

### FOLDED STIFFENERS

2 strips, 28 x 3 cm, across the lower bridge; 4 strips, 16 x 3 cm, along handles. Fold lengthwise.

### THROTTLE BOX + SLOT

Shoebox about 30 x 16 x 12 cm. Cut a 20 x 1.2 cm slot in the lid, starting 5 cm from each short end.

### SLIDER + GUIDE RAILS

Carriage: 10 x 8 cm. Rails: 2 strips, 26 x 2 cm. Fold and mount 8.5 cm apart under the lid.

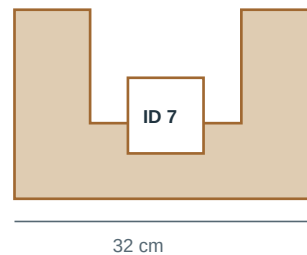
### THROTTLE MARKER PLATE

1 piece, 8 x 8 cm, on an upright bracket fixed to the moving carriage.

### OPTIONAL YOKE REST

2 sides, 16 x 12 cm, with 3 cm-deep top notches; base, 22 x 16 cm. Stand sides 12 cm apart.

## 02 ASSEMBLE THE CONTROLS



### U-shaped yoke

20 cm tall  
Opening:  
16 x 12 cm

Diagram is not a cutting template.

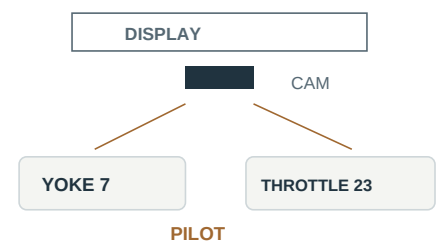
**Yoke:** laminate the two U shapes with the cardboard flutes crossing. Add folded ribs and tape the handles. Mount the **70 mm ID 7** marker flat; keep its white border clear. Rest wrists on the desk to reduce wobble. The optional cradle is for parking: lift the yoke free to roll and tilt.

**Throttle:** slide the carriage between the rails. Tape a capped pen upright through the slot as the grip; add an underside crosspiece so it cannot pull through. Put the **50 mm ID 23** plate just in front of the grip, facing the camera and moving with the carriage.

Add end stops and paper shims for gentle friction. Label near end **IDLE**, middle **CRUISE**, far end **FULL**. Tape the box to the desk. Keep fingers and glossy tape off both markers.

## 03 POSITION THE CAMERA

Mount it **70-100 cm in front** and **25-40 cm above the desk**, aimed down slightly. Both marker faces point toward it. Raise or offset the camera until throttle travel visibly moves in the preview. Use diffuse light. Keep spare marker copies out of view.



## 04 CALIBRATE, CENTER, FLY

Run `.venv/bin/python vision/tracker.py --camera 0 --calibrate`. In the tracker window, hold each pose for one second and press **Space**: neutral, left, right, forward, back, idle, full. Center the yoke, idle the throttle, select vision in the game, then start. Recalibrate after anything moves.